

Supporting International Trainee Teachers of Physics:

Guidance for International Teachers

DRAFT

Foreword

Example subheading

To be written

Introduction

In recent years there has been a significant increase in the number of international candidates undertaking initial teacher education (ITE) in physics across England, increasing from 23 in 2022/23 to 242 in 2024/25.

There is also a rise in the number of qualified physics teachers moving to England to teach in schools. These new trainees and qualified physics teachers are helping to address the ongoing national shortage of secondary physics teachers. This guidance is primarily aimed at international trainee physics teachers who are on a university-based course, but it will also be useful to support those on other programmes and international teachers who are entering directly into employment in UK schools. It sits alongside some parallel guidance and advice for others involved in supporting international physics trainee teachers e.g. ITE tutors, school mentors, school leaders. This guidance has emerged from a collaboration of science and physics teacher educators from different teacher training providers and the Institute of Physics, who have sought to learn from the experiences of international physics trainee teachers they have worked with over the past two years. Their practice as teacher educators and mentors has been enriched and improved through engaging with and listening to this diverse cohort of new, resilient and enthusiastic trainee physics teachers. They also feel encouraged about the positive impact that this group of physics teachers will have on the pupils they teach in the future. Recognising that there are some distinctive issues facing international trainee teachers, the intention of this document is to offer advice to support this cohort acclimatise to the demands of the English education system.

This guidance is grouped into 5 sections:

- Before you apply
- Before you travel
- On arrival
- Thriving through your training

- Applying for jobs

Before you apply

Understand the structure of the teacher training course

Teacher training is an intensive course that runs from September to June. Successful completion of teacher training leads to Qualified Teacher Status (QTS) which enables you to teach in state schools in England. The structure of the teacher training course varies depending on your provider, but most of your time will be spent on placement in two different schools, perhaps interspersed with blocks of time in your teacher training provider. The school year is divided into 3 “terms” which run from September to December, January to April, and April to July. Schools typically have a half-term break of between 1 and 2 weeks in October, February and May. It is likely you will attend sessions in your training establishment at this time. You will have a two-week holiday at Christmas and at Easter which coincides with the state school holidays. If you are on placement in a private school or an academy, the holidays may be extended.

When you start a teacher training course, the first few weeks will be spent understanding some of the key themes of teaching, such as assessment, pedagogy, inclusion and behaviour management. It is likely you will also have a personal tutor with whom you may be able to discuss your experiences so far and any issues or concerns you have regarding the course. Forming positive relationships with your academic tutors and peers on your course is very important as these will provide you with support networks during your time as a trainee teacher.

Assessment of your progress is typically made up of school practice reports from your placement schools which highlight your key strengths and targets for improvement as well as a number of Postgraduate/ Master’s level assignments. These may consist of independent or group presentations and individual assignments. To complete the course successfully, you will need to show that you fulfil the [Teacher Standards](#) from evidence in your school practice as well as pass any other assessments that may be set by your provider. Your progress is reviewed regularly during your school placement. Each institution will have minimum requirements on the progress you make. Failure to satisfy these requirements may lead to your withdrawal from the course.

Understand the structure of the teaching placements

You will be training in secondary schools which are for children aged 11 –16, or 11 -18 if the school includes a sixth form (covers age 16-19 education). The school week runs from Monday – Friday in state schools. The day generally starts between 8.30 and 9:00 and finishes between 3.00 - 4.00pm. Qualified teachers with no additional responsibilities teach over 20 hours per week so much of the preparation, marking and administration (admin), is completed out of school hours and therefore teachers typically arrive much earlier and stay later to complete this. You may also be expected to stay later to attend department meetings, training sessions and parents’ evenings on specific days.

There are typically 5 teaching hours / per day, which are split into various lesson timings in a strict timetable according to the specific school. The timetable may operate on a one or two-week structure. There is normally a short morning break, a longer lunch break and registration/pastoral session often called ‘form time’ (or sessions in the morning and afternoon). Teachers typically have pastoral responsibility for their form (or tutor group). It is common to teach approximately 30 pupils in each class. Secondary education is compulsory for all, and attitudes to learning vary greatly. Science is a core subject meaning all pupils are required to study it. Pupils may be placed in classes according to their prior attainment. This is called setting. If pupils are placed in groups according to their attainment in a single subject. Pupils can also be placed into broader groups for all subjects based on their general attainment in all subjects. This is called streaming. Setting and streaming are more common in years 10 and 11, which are also known as Key Stage 4 (KS4), when pupils study for GCSE examinations. It may be likely however that you will be teaching mixed attainment groups. Teachers must therefore adapt their teaching to stretch and challenge the higher achievers while ensuring that pupils with the lowest prior attainment can access and understand the material.

You will be provided with a designated mentor in each placement school who will be responsible for supporting and guiding you on placement. You will have a weekly meeting with the mentor to discuss your progress, successes and targets for improvement. In school you will gradually build up your teaching hours in collaboration with your school mentor, depending on your confidence and previous experience. You are likely to start by initially observing lessons followed by gradually increasing your teaching hours to the timetable specified by your ITE provider, the total teaching hours will vary per provider. You will be expected to write a full lesson plan and a self-reflective evaluation for a number of lessons per week as specified by your ITE provider. It is likely you will be required to hand in the lesson plan to your mentor in advance (normally 24 or 48 hours) of the lesson so they can provide you with constructive feedback to improve your delivery.

Understanding importance of safeguarding

Safeguarding relates to the right of children to be protected from harm. This is the primary responsibility of all teachers and as part of your training you will have compulsory safeguarding training with your ITE provider and placement schools. Schools have a key role to play by ensuring they have policies in place to ensure children are protected and identify children at risk. All trainee teachers are expected to comply with safeguarding rules at all times. Make sure you are familiar with policies in your placement schools when you begin the placement. We also recommend that before starting teacher training, all applicants make themselves aware of statutory guidance, Keeping Children Safe in Education 2024: <https://www.gov.uk/government/publications/keeping-children-safe-in-education-2>

Understand the demands of the course

Teacher training is intensive, full-time course, with high levels of contact time. You will be required to juggle a teacher's workload including school-based tasks such as planning lessons, teaching and marking alongside the demands of completing several Master's level academic assignments. This requires a high degree of organisation and flexibility. The nature of the course is unlike other postgraduate qualifications. It is important to understand that successfully completing this course while working part-time is extremely challenging.

Before you travel

Explore visa options

Most international trainee teachers will need to obtain a student visa before commencing training. To ensure you are able to travel and begin the training on time it is essential to apply for your student visa as soon as possible. Upon successful completion of your teacher training programme, you will need to switch to a different visa type which will probably be either a graduate visa or skilled worker visa. The type of visa will depend on gaining sponsorship from an employing school. It is important to be fully informed of the advantages and disadvantages of each option as well as the cost implications for both you and your future employer.

Ensure you understand the financial Implications of your teacher training course

The Department for Education provides generous bursaries to train to teach physics in England . These bursaries are valuable to cover the costs of studying in England, however the cost of living can be high. You will need to pay costs such as university fees, accommodation, food, utility bills, travel and budget for your own entertainment. Some providers may also have other funding available that you can apply for when studying. Please see the link below to compare the cost of living in different cities in England to other cities in the world: https://www.numbeo.com/cost-of-living/country_result.jsp?country=United+Kingdom Be aware of fluctuating exchange rates when transferring funds from overseas. You will not receive your first instalment of the bursary until the end of October at the earliest but will have significant expenses before this, so you need to make sure you have sufficient funds available on arrival in England.

Research accommodation options

If possible, try to arrange accommodation before arriving in the UK, even if it is only on a temporary basis while you search for something more permanent. Having a UK address is important, as it allows you to set up a bank account and apply for a DBS check (see section below).

Accommodation in England can be very expensive, especially in large cities such as London. You are often required to have a reference to confirm you have sufficient funds to pay the rent.

Choosing accommodation close to good transport links and within a reasonable journey time from your training provider will save you time and money. Landlords may ask you to rent the property for a full year or 48 weeks and often require deposits and a guarantor reference to secure tenancy. Gather any documents that may be required to secure a tenancy – such as a guarantor reference or a bank statement to evidence proof of funds prior to arrival.

University accommodation, where available, will not usually be accessible until mid to late September so you will need to secure alternative housing for the interim. Some teacher training providers may have housing advice services for international students that provide opportunities to rent cheaper rooms in shared houses or flats. When considering accommodation aimed at students, be aware that the dates of term for teacher training programmes are often different from other university courses – make sure that your rental agreement will allow you to live in the property from the very start to the end of your course.

Review subject knowledge

Science teachers in England generally teach Physics, Chemistry and Biology regardless of their specialism. You will be expected to have secure knowledge of all three sciences covering the English [KS3](#) (11 -14 years) and [KS4](#) (14 – 16 years) curriculum. Take the opportunity to refresh and update your subject knowledge before you travel, as you will have limited time to do this once the course begins.

Review your digital skills

Computers are essential tools for teaching in England, and it is highly advisable that you have your own laptop. Lesson content is often presented in slideshows; communication between teachers is by email; grades are recorded, and data is analysed in spreadsheets; resources are stored in shared drives. You will be expected to use common applications with confidence and proficiency. Ensure your skills are up to date before you start the course, the MS Office suite of apps including Outlook, Teams, PowerPoint and Excel, are most commonly used but some schools use alternative platforms such as Google.

Familiarise yourself with the local dialect

The way English is spoken varies greatly across the country. Even if you are fluent in English, local accents, dialects and slang can be hard to understand, especially when spoken quickly. It can be very demanding making sense of what is said, and you may feel extra tiredness at first. Once you know where you will be based in England, try to find TV programmes or films set in the region and watch with subtitles so you can tune your ear to the local language as much as possible before you arrive (Be aware that you need a licence to watch TV in England: <https://www.gov.uk/find-licences/tv-licence>). When you arrive, encourage the native speakers and your mentor to correct your pronunciations and written English in lessons.

On arrival

Set up a bank account

It is essential to set up a bank account promptly, as you may need one in place before you can rent accommodation or apply for a DBS (see below). Banks will require proof of identity and your

address. If you are registered as a student, you may be exempt from Council Tax, which is a local tax based on the value of the property you are occupying. Many organisations and companies give discounts and offer deals for students. This includes some banks, so it is important to shop around to get the best deal for you.

Apply for DBS Certificate (Disclosure and Barring Service)

To work with young people in England a DBS check is necessary. Many providers will have links to information regarding the application process. It is likely you will be sent a link by your ITE provider to initiate the process in early summertime. You will need a home address in the UK to complete your application – it is essential you add your UK address as your postal address on the application form rather than your address from back home. It takes at least 2 weeks to receive your DBS certificate once your application has been submitted, however be aware that the process can be much longer. The certificate is required to start your first school placement as you may need to provide the physical copy of your DBS to the school. Delays can therefore affect your engagement with your placements in training schools.

Orientate Yourself

Many ITE providers run pre-course sessions either in-person or online. Make sure you find out where and when these take place and ensure you attend as the sessions will be a useful source of information as well as providing you with opportunities to socialise with colleagues. Find out about wellbeing services that are offered.

It can be very daunting when you first arrive at a new institution so ensure you arrive early and ask for a map of the campus to locate the key rooms you will be using. Find out who the key student contacts are e.g. your course leader and student service advisors and ask if you are unsure.

Take time to explore your provider's website and complete any pre-course reading. It is also useful to look over your timetable, so you know when and where you need to be. Punctuality and attendance are very important. Attendance at the start is vital in terms of receiving the relevant information and ensuring you are organised and feel confident and in control.

Thriving through your Training year

i). Settling into life in the UK

Everybody has different feelings and challenges when moving to a new country. It is perfectly natural to feel overwhelmed and excited about your new situation at the same time. However, there are some common elements which you may face and there are strategies you can adopt and approaches you can consider to adjust to your new environment.

Access Healthcare

You will have access to free medical treatment (GP and hospital service) through your annual NHS surcharge (included in your visa). You will however need to pay to visit a dentist, optician and also any prescriptions. It is advisable to register with a GP (General Practitioner) practice close to where you live to be able to access free consultations.

Explore Transportation options

Use transport websites and apps to find the quickest/cheapest way of travelling. Some buses or trains may be cashless, and you may need to set up a payment card or app to use. Some transport providers also have day and weekly travel cards allowing you to commute for a set fee. There may also be student discounts. Also consider walking or cycling! If there is more than one trainee teacher or another teacher at the school who lives nearby and has a car you could consider sharing lifts to share the costs of fuel.

Prepare for the English Weather

It is likely that you will arrive in England at the end of the summer when the weather is quite mild (15 to 20°C). Remember that England is located at about 52 degrees of latitude hence subject to seasons and varying daylight time. Hence around the summer solstice, the days are very long, and you can enjoy long evenings until 10 pm. However, temperatures drop significantly over the Autumn months with snow likely in the north and higher regions in winter. Do not underestimate how cold, wet and dark it can be in winter, especially in northern areas when temperatures are typically 0 to 10°C but frequently fall below freezing. Investing in waterproofs, umbrella and warm jumpers, scarves, hats, gloves and a hot water bottle at the beginning of the course is advisable. The weather

changes daily and often during the day itself so it is useful to check the weather forecast and ensure you wear warm clothes, and weatherproof shoes as necessary. If you are used to a warmer and lighter climate you may find it helpful to research seasonal affective disorder (SAD) and ways of mitigating it e.g. with light therapy.

Understand the importance of Punctuality and Time Keeping

Being on time is a cultural expectation in England. This applies to all parts of life but especially school and ITE life. Most schools and teacher training institutions have set time of days that staff need to arrive by, so you need to ensure you allow sufficient time to travel, and a bit extra to allow for any unexpected delays. Schools have high standards of attendance and punctuality for their pupils. Teachers act as role models so you need to take proactive steps to ensure punctuality. For safeguarding reasons, most schools do not allow the use of mobile phones by either staff or pupils. It therefore is useful to own and use a wristwatch in order to track time during lessons and help with punctuality.

Find your national foods

In many places it is possible to find a shop that sells products from around the world. You may find this a source of comfort and familiarity. There are often online communities on social media sites that can recommend specific shops or restaurants. Some universities have student led societies representing different nationalities or cultures who also may be able to help find familiar foods.

Shop on a budget

Make sure you familiarize yourself with the different supermarket brands to find the best value for money on essential products. As well as selling food, large supermarkets often sell a range of good value clothing and household items. Second hand charity shops resell pre-owned electrical items, furniture and clothes cheaply (e.g. Cancer Research, Oxfam, British Heart Foundation, Sue Ryder). Recommendations from local people may be the most beneficial way to shop.

Dress Code

Schools in England often have formal dress codes. For men, this means a suit with shirt, tie and smart shoes is expected. For women this means equivalent business attire. Culturally in England it is better to be overdressed than dressed too casually until you become familiar with the culture of the particular school. If you have questions about appropriate dress for school, these are best asked to your school mentor. In contrast to schools, in most universities, students dress casually for taught sessions.

Find a Support Network

Many towns and cities have vibrant communities of people speaking different languages. If you are religious, you may find support through faith-based organisations or, if you are a student at a university there may be student societies you can join that celebrate different cultures. It is good to find community and support outside of the teaching environment! You may feel overwhelmed when you arrive and experience homesickness and feelings of isolation. Make sure you talk to your personal tutor about your feelings and concerns. Also, take opportunities to start conversations with people. Your colleagues will also become an important support network. Keep active and joining a gym club or sports clubs are also useful for developing social networks.

Deal with Separation From Loved Ones

Relationships are very important and being separated from loved ones is difficult. It can impact how you feel and affect your progress when you are training to teach. Most mentors and tutors will be supportive and try to help if they know and understand the issues you are facing. Beat the homesickness by printing photos of your loved ones, sharing a regular communication time with significant others. If you need free internet to video call, universities may be able to provide this.

ii). Settling into teaching in the UK

Even if you have extensive teaching experience in your home country, it is likely that you will find teaching in the UK a very different experience. Every school is different, so be prepared to meet different structures and practices in different placements, but there are many aspects that are common to all.

Make a Good First Impression

First impressions count! On the first day plan your journey and allow for contingency to ensure you arrive on time. Keep a positive attitude, smile a lot, apologise if you make a mistake and express enthusiasm about the school and your place in it. Remember that you are new to the environment and you may find it overwhelming to start with. High standards of professionalism are expected of all staff in schools including new trainee teachers. It is best to avoid voicing negative opinions or concerns about practices in the school because this may be seen as disrespectful. If you have questions, the best thing to do is to write them down and then discretely find a time to politely consult your school mentor or university tutor.

English people have a reputation for being overly polite and expect politeness from others, e.g. holding doors open for others, saying please, thank you and sorry often.

Prepare for School Placements

School placements are normally arranged by your provider. You will have to plan your journey to your school and will normally have at least two different placements. Schools do not offer accommodation for teachers or provide transportation.

Even if you have taught before it is important to realise that all schools are different and will have different policies and requirements, ranging from ways of teaching, working hours and marking standards for example. It is likely there will be significant differences to schools in other countries in terms of culture, curriculum and professional standards (for example, the amount of teacher talk in lessons, the ways in which teachers relate to the children they teach...). You might find that a major difference is in the amount of school policies and the expectation that these are adhered to in areas such as lesson planning, assessment, pupil behaviour management and safeguarding. A key skill required of teachers is that they are adaptable and ready to learn from the expert teachers in your school!

Schools value members of their community from all cultures and so often will have accommodation for religious observance such as prayer rooms and will normally also allow leave to be taken for religious festivals. These arrangements should be discussed with your school mentor.

Learn about your school via their website as soon as you know where your placement is. The information on school websites is aimed at parents rather than teachers, but you can read policies, identify key members of staff, get a feel for the school culture and find out about how the curriculum and pastoral system are structured. When you arrive, carry a notebook with you as you will be given a

lot of information. Don't worry if you find the amount of information overwhelming to start with, schools can be quite difficult places to understand.

Don't forget to ask about basic practicalities such as where staffrooms are and which toilets you should use.

Introduce yourself to pupils

In England teachers are generally referred to by their title and surname in front of pupils, but by their first names with colleagues. When introducing yourself to pupils you should therefore say you are "Mr ... or Mrs ... or Mx ...". All women are normally called 'Miss' no matter whether they are married. Men are usually referred to by pupils as 'Sir'. Teachers who identify as non-binary may be referred to as Mx (pronounced 'mix').

Ensuring your speech is understood

You may feel worried about pupils understanding your accent when you teach. This may not be as significant as you might think. Many schools are a diverse community of learners and teachers are used to listening to people who speak in different ways. However, some areas of the UK are less culturally diverse and it may more take time for you to feel comfortable. To gain confidence, when you introduce yourself to a new class you could confront the issue head on and make it clear to the pupils that they should ask you if there is something you have said that they don't understand. You might also consider practicing giving key explanations or instructions with a partner or mentor who could help you pronounce specific scientific vocabulary. You could also record yourself giving explanations or instructions, then play it back and compare to a native English speaker.

Identifying pupils by name and face

The ability to use pupils' names is an essential way of building positive relationships and managing behaviour. A strategy to help is to use seating plans. A seating plan is a document showing the layout of the room and where each pupil sits with their name. Seating plans may also include photographs (don't take your own photographs, schools will have them already). Class teachers may already use seating plans and you may find it helpful to discuss why children have been placed in particular seats. Check and practice the pronunciation of pupils' names before using them. A good idea for starting to use names is to greet each pupil as they enter the class and say goodbye as they leave.

Try to speak to each child individually during a lesson, for example by directing questions making a point of using their name. It is normal for pupils to be registered to confirm they are in each class (and a legal requirement each morning and afternoon). Teach the pupils how you would like them to answer when you call their name out. Let them know how you would like them to respond if you make a mistake when pronouncing their name.

Managing Pupil Behaviour

It is essential to read your placement school's behaviour management policy. Note down how you should reward positive behaviour and how pupils are sanctioned, step by step. The school policy will normally have a system of warnings and escalation of consequences. The majority of pupil behaviour issues are 'low level disruption' which means things like interrupting the teacher, talking when they should be silent, disturbing others or repeatedly not following instructions. Other common issues are dealing with lateness, homework, equipment and uniform problems. Different schools often take different approaches to managing pupil behaviour so don't assume that the system in your first school placement will be the same in your second.

Building relationships with pupils is key to managing behaviour. Teachers are encouraged to challenge poor behaviour in ways that do not humiliate or belittle pupils. Teachers develop a highly skilled use of language and non-verbal communication to direct behaviour. It is normal for all teachers to practise behaviour management throughout their careers and it will take time to hone your own skills. It will be necessary to closely observe expert teachers and use their practice as a model for building your own approach.

Corporal punishment of any kind is totally prohibited in England. Physical contact with pupils should be avoided unless there is serious risk to the safety of a pupil or staff member.

Developing Inclusive Teacher Practices

Science lessons in school in England are compulsory for children up to the age of 16 (Year 11). Therefore, pupils have varying levels of motivation and engagement. It is important that you engage with all pupils in your lessons and your lessons are accessible and inclusive of all pupils. The IOP's [top tips for inclusive science teaching](#) are a useful guide. You will teach some pupils with special educational needs (SEND), some of which will have an Educational Health Care Plan (EHCP) that provides useful teaching strategies to support the pupils. Teachers usually have access to a simplified version called a learning plan or pupil passport, which advises teachers on the best

approaches to use with a particular pupil. You may also have pupils for whom English is an additional language (EAL) who will also need support with literacy.

Many people in the England identify as being part of the LGBTQIA+ community. This includes people who come under the umbrella of lesbian, gay, bisexual, transgender, queer/ questioning, intersex and asexual /aromantic. It is highly likely that you will have colleagues, pupils and parents who are part of this community whether openly so or not. Homophobia and transphobia are not acceptable and must be challenged.

It is important to note that in England, families can take all shapes and sizes. Children may be brought up by a single parent, a same-sex couple (two mums or two dads), a mixed sex couple (mum and dad) or another member of the family (e.g. aunt, uncle, grandparent). They may be adopted by another family that is not their birth family. All families are celebrated. Some children are looked after by foster families or live in children's homes; they are often described as "cared for". Not everyone will go on to make their own family – some people choose not to have children. You will also notice the term 'guardian', being used instead of parent/father/mother.

Be careful not to make assumptions based on your cultural norms. Comments on someone's sex, sexual orientation, race, religion, age, disability or family relationships could be misinterpreted and are best avoided. When referring to pupils' guardians you may find it useful to adopt the phrase 'to your parents, carers or whoever is at home'.

Preparing to Teach Sciences

It is common in England for physics teachers to teach all three sciences, especially to younger pupils, and often at GCSE. This may involve additional study of the curriculum and strengthening your subject knowledge in the other sciences. Find out who is in charge of biology and chemistry as well as physics in the school and seek help where necessary.

Normally when teaching older pupils you will focus on your subject specialism of physics.

Schools use "schemes of work" that show what topics need to be taught to each class when, and may include an outline of what is to be covered each lesson, but the detail varies in different schools. It is important to keep to the schedule of lesson planned out by the school. This is because there are often tight time frames for completing topics to prepare for assessment and multiple parallel classes studying the same course.

Most classrooms have projectors and screens or interactive whiteboards. Teachers often use PowerPoint presentations to present information. In some schools, slides are shared and in others, teachers write their own. You will be expected to prepare or adapt lesson presentations however lessons should include a variety of other learning activities. You will also be expected to write lesson plans. It is normal to be required to submit lesson plans well in advance of teaching (for example 48 hours).

Reviewing Your Physics Subject Knowledge

Most well qualified physics teachers find that there are aspects of the school curriculum that they are less familiar with. A part of being a professional physics teacher is to recognise where you have gaps in knowledge or misconceptions and address these. Finding ways to explain complex physics ideas in simple ways is a highly valued skill you must develop. In England the focus is on understanding rather than giving definitions and manipulating equations. This can be challenging if you are not used to it! You may find it helpful to script and practise explanations with a peer or mentor before using them in the class.

[Physics CPD Videos | IOPSpark](#) are a good starting point to check subject knowledge and teaching methodology

Mathematics skills are integral to Physics and form 30% of the GCSE exam content (and 40% at A level). It is an area where many pupils lack confidence and can often be a source of anxiety. It is worth liaising with the maths department to ensure that you are consistent in methods and notations so as to provide a coherent approach. See “Make time for maths” in the IOP’s [Top Tips for Inclusive Science Teaching](#)

Developing Physics Specific Pedagogy: e.g. Key terminology, practical work, misconceptions

Physics in the school curriculum is largely based around pupils understanding of scientific concepts in real life contexts. Practical investigative work is an integral part of learning all topics. Pupils are encouraged to ask questions and be curious about the world. There are often key words that must be learned by pupils to explain concepts. You need to become confident in leading pupil practical work and demonstrations, so practice is essential (see section below regarding Science technicians).

in school will often help if you ask. The IOP Spark website's [Practical Physics | IOPSpark](#) section contains a large collection of resources for practical work including methods, safety guidance and teaching notes. Pay particular attention to how other teachers give guidance on aspects of practical work such as rules for drawing graphs and tables and how variables are classified.

Every pupil arrives at each lesson with their own life experience of science and may well have formed their own explanation of scientific phenomena, many of which may be incorrect, such as they may think that a force is needed to keep an object moving. These erroneous ideas are known as misconceptions (also called pre-conceptions, alternative conceptions). It is crucial to familiarise yourself with the common misconceptions in a topic when lesson planning, and don't be surprised to find that you may hold some of these ideas yourself! Most of the common misconceptions are discussed on IOP Spark [Search | IOPSpark](#) where you can search by topic and pupil age group.

It can be challenging to pitch lessons at the right level for each class when beginning to teach. It can be easy to assume too much previous knowledge from pupils, particularly in their comprehension of scientific vocabulary. Some things that may help are to use previous learning outcomes, discuss prior learning of your class with other teachers, and plan formative assessment activities in your lessons.

When observing other teachers keep a record book with you to note down key terminology, teacher phrases and contextualised examples used.

Working With Science Technicians

Most schools have professional science technicians whose role is to support pupil learning by providing expertise and resources for practical work. Technicians are highly skilled and should be consulted about any practical work you plan for your lessons. They are also a helpful source of advice if you are unsure about health and safety or a particular method for an experiment. It is normal that any equipment you require for a lesson is requested in advance – usually a week. Check the process for ordering equipment in school and always be on time!

Risk Assessment for Practical Work

It is important that physics teachers know how to manage practical work safely in the classroom. This can raise complex issues (e.g. when demonstrating radioactive sources or using electrical current). A good source of guidance is the organisation CLEAPPS who provide information to science teachers to help them risk assess activities: <https://www.cleapss.org.uk/>. CLEAPSS has a large bank of practical work resources and advice alongside the information required for risk assessments. Every school should have log in details. You can sign up to receive regular newsletters.

You are expected to risk assess every practical and are responsible for the safety of the pupils in your class. There will be standard lab safety rules in your school that must be followed, but each activity is different. If you are unsure, you should consult an experienced member of staff or lab technician.

Communication in Schools

Most schools are organised using IT systems and teachers are expected to engage throughout every day with different digital platforms. Teachers can also access shared resources from a departmental drive. If you would like to download and save documents from the school's system, you must ask permission first.

Normally it is expected that teachers will check their email for urgent messages regularly throughout each working day. Online collaboration tools may be used (such as Microsoft Teams) to send messages or documents. Pupil attendance will usually be taken using an online software and most schools will communicate and share learning with pupils via an online learning platform. It is very important that you learn how to use all these systems and if you have problems to get them resolved quickly – schools will have IT support or a technician you can consult. Don't be afraid to ask for help. Some schools have 'pigeon holes' which is a box for each person in the staff room where paper communication or mail can be left. If these are present it is expected that teachers check these daily.

Teaching Unions

Most teachers in England belong to a union. As well as offering collective bargaining on issues such as pay and working conditions, they will provide legal support should you need it.

Applying for Jobs

The job prospects for physics teachers in England are very good! Providing you successfully complete your teacher training you will gain Qualified Teacher Status (QTS). Many ITE providers will offer help and advice on applying for jobs and may also share details of vacancies arising in their partnerships with schools. You may also ask at your placement schools while you are training (if you are enjoying working there) if they will have any vacancies.

In England, teaching jobs are primarily advertised on 2 websites: teachingvacancies.service.gov.uk and www.tes.com/jobs. Schools also advertise vacancies on their school websites and sometimes welcome direct applications. It is worth checking therefore if there are particular schools you would like to work at. It is crucial that you take time to adjust applications to specific schools and job descriptions using the school website as a useful source of information.

When you get your first teaching job you will be an 'ECT' or Early Career Teacher for the first two years. This means you are entitled to a slightly reduced teaching load and bespoke professional development. You should look for jobs that advertise as being open to ECT's and look out for information on the further training opportunities they can provide for you. As you are in your training year you will usually be looking for jobs that start in the September after you complete your course. These will mainly start to be advertised in January onwards with the largest number of jobs advertised being around April.

Before applying, you might find it useful to check if the school has sponsored international teachers in the past. This Home Office website provides a list of schools that have done so.

<https://www.gov.uk/government/publications/register-of-licensed-sponsors-workers>

There is a shortage of physics teachers in schools, therefore it is always valuable to contact the school and discuss the possibility of sponsorship before you apply even if the school is not on this list. Alternatively, Visa sponsorship can be discussed in your interview.

If Visa sponsorship is not possible, ensure you meet with a Visa compliance officer (or equivalent) at your ITE provider to talk through alternative options eg a school may consider starting you on a higher salary to cover the costs of a Graduate Visa.

If you are successful at interview a headteacher will normally make a job offer on the same day. During the day it is normal to be asked “Are you a serious candidate?” or “If we were to offer you the job would you accept?” There can be some pressure to accept a job offer there and then, so if you are unsure at all you could ask for a couple of days to discuss with family and friends before accepting.

Don't be disheartened if you don't get jobs straight away, it may take a number of interviews. It is important to see all your interviews as part of a learning process. Many interviews may require you to teach a lesson. Always ask for feedback on your performance and proactively embed this into your everyday practice in your placement school. Working collaboratively with your mentor to cement your teaching expertise is the key to your long-term success in securing a role. Try to avoid attending numerous interviews in succession as this will reduce your time on placement which may have repercussions in terms of an extended placement.

Common Acronyms and Abbreviations Used in Schools

Acronym	Short for...	Meaning
ADHD	Attention Deficit Hyperactivity Disorder	A commonly diagnosed neurodevelopmental condition in which people may struggle to concentrate and be impulsive.
ASD	Autism Spectrum Disorder	A commonly diagnosed neurodevelopmental condition in which people may find social interaction difficult but excel in other ways.
AfL	Assessment for Learning	An approach to pedagogy that uses formative assessment to provide regular feedback to pupils and also inform teaching.
AP	Assistant Principal	A senior member of staff, just under Vice Principal in the school hierarchy.

ASE	Association for Science Education	A membership based charity that supports science education.
CLEAPSS	Consortium of Local Education Authorities for the Provision of Science Services	A UK-based advisory service that provides health and safety advice for practical work in science and design and technology lessons – all schools should be members. (www.cleapss.org.uk)
COSHH	Control of Substances Hazardous to Health	The law relating to the control of hazardous substances.
CPD	Continuing Professional Development	Any training or professional development after ITE.
DBS	Disclosure and Barring Service	A UK government service that helps employers make safer recruitment decisions. It issues DBS checks to reveal if a person has a criminal record or is barred from certain types of work. The aim is to assess suitability for working with children and vulnerable adults.
DfE	Department for Education	Government department in charge of education.
EAL	English as an Additional Language	Describes children for whom English is not their mother tongue.
ECT	Early Career Teacher	The first two years as a qualified teacher.
EHCP	Educational Health Care Plan	A document about children with SEND that outlines their needs and provisions for those needs.
FSM	Free School Meals	Children who are from low-income backgrounds are entitled to a free meal

		at lunchtime when in school. This term is often used to classify children from a disadvantaged background.
GCSE	General Certificate of Secondary Education	The main academic qualifications, mostly assessed by examinations at age 16.
G&T	Gifted and Talented	Children who are high attainers or have the potential to be high attainers.
HLTA	Higher Level Teaching Assistant	Not a fully qualified teacher but someone who is qualified to lead lessons.
HoD	Head of Department	The teacher with responsibility for a specific subject area in a school.
HoY	Head of Year	A teacher with pastoral responsibility for a year group in a school.
HEI	Higher Education Institution	An organisation such as university offering degree level and post-graduate qualifications.
IB	International Baccalaureate	A post-16 qualification in which students study a broader range of subject that they do at A level.
ICT	Information and Communications Technology	A general term for the use of computers and digital technology.
INSET	In-Service Training Day	A day dedicated to staff training when pupils are not in school.
IOP	Institute of Physics	The learned body for Physics in the UK who provide lots of teaching resources.
ITAP	Intensive Training and Practice	Are periods of the ITE programme which focus intensely on specific aspects of teaching. They are intended to help

		consolidate knowledge of key evidence-based principles for effective teaching, provide time for practice and in lessons and get feedback.
ITE	Initial Teacher Education	Teacher training prior to qualifying as a teacher.
ITTECF	Initial Teacher Training and Early Career Framework	The statutory curriculum for the initial teacher training course and the early career support that new teachers are entitled to.
KS3	Key Stage 3	A phase of secondary education in the United Kingdom for pupils aged 11-14. It is designed to build on the knowledge and skills gained in primary school (Key Stage 2). This typically covers years 7-9 in England, Wales and Northern Ireland.
KS4	Key Stage 4	A phase of secondary education in the United Kingdom for pupils aged 14-16, usually covering years 10 and 11. Students are preparing to take GCSE examinations or equivalents.
KS5	Key Stage 5	Sometimes referred to as 'post-16' or 'Sixth Form'. Education in years 12 and 13 for 16-18 year olds, culminating in qualifications such as A levels.
LSA	Learning Support Assistant	A member of staff who comes into lessons to support pupils with SEND.
MPS	Main Pay Scale	The nationally agreed pay scale for teachers in their first 6 years.

NASUWT	National Association of Schoolmasters and Union of Women Teachers	The second largest education union in the UK.
NEU	National Education Union	The largest education union in the UK.
OFSTED	Office for Standards in Education, Children's Services and Skills	The government body who conducts school inspections.
PCK	Pedagogical Content Knowledge	The integration of subject matter expertise and teaching strategies.
PP	Pupil Premium	Additional funding provided to schools by the government to support disadvantaged pupils. For example, those on FSM or with parents in the armed forces.
QTS	Qualified Teacher Status	The professional qualification normally required for a teacher to gain employment in a state school in England.
RSB	Royal Society of Biology	The learned body for Biology in the UK.
RSC	Royal Society of Chemistry	The learned body for Chemistry in the UK.
SAPs	Science and Plants for Schools	An organisation supporting the teaching of plant science.
SCITT	School Centred Initial Teacher Training	Teacher training programmes that are based in schools rather than HEIs.
SEND	Special Educational Needs and Disabilities	A term used for children who have been identified as having disabilities or needing support with learning.
SENCO	Special Educational Needs and/or Disabilities Co-ordinator	A teacher who leads the school's provision for children with SEND.

SEMH	Social, Emotional and Mental Health Difficulties	A broad term to define a range of different needs children may have.
SK	Subject Knowledge	The understanding of the concepts, principles, and facts within a specific discipline.
SKE	Subject Knowledge Enhancement	The improvement of subject knowledge usually through a subject specific training course taken prior to starting teacher training.
SLT	Senior Leadership Team	This includes the Head Teacher or Principal, Deputy Heads or Vice Principals, and Assistant Heads or Principals who oversee the running of a school.
SPaG	Spelling, punctuation and grammar	This covers precise and correct use of the English language.
SoW / SoL	Scheme of Work / Scheme of Learning	A document outlining what is to be taught in each topic. Normally includes lesson by lesson breakdowns of learning objectives and learning activities.
SSR	School Science Review	A magazine dedicated to school science education published by the ASE.
TA	Teaching Assistant	Someone who supports a teacher in the classroom. TA's help individual students or small groups understand lessons, often including those with SEND.
TLR	Teaching and Learning Responsibility	Specific responsibilities of teachers above and beyond their own classroom teaching, that entitle them to extra pay.

UPS	Upper Pay Scale	A higher pay scale than MPS that can be accessed by teachers with more than 6 years' experience.
VA	Value Added	A measure of how much progress a child makes, usually from one Key Stage to the next.
VP	Vice Principal	Equivalent to a Deputy Head Teacher. The most senior teachers apart from the Principal or Head Teacher.